

Solutions to Problems 9.3 and 9.4

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Problem 9.3: Rayleigh Number for Various Fluids

Given: characteristic length $L = 0.01$ m, temperature difference $\Delta T = 30$ K. Rayleigh number

$$\text{Ra} = \frac{g \beta \Delta T L^3}{\nu \alpha} = \text{Gr} \cdot \text{Pr},$$

with $g = 9.81$ m/s², volumetric expansion coefficient β , kinematic viscosity ν , and thermal diffusivity α . Using representative properties at the stated conditions:

Fluid (state)	ν (m ² /s)	α (m ² /s)	β (K ⁻¹)	Ra
Air (1 atm, 400 K)	2.83×10^{-5}	4.50×10^{-5}	2.5×10^{-3}	5.8×10^2
Helium (1 atm, 400 K)	1.72×10^{-4}	2.69×10^{-4}	2.5×10^{-3}	1.6×10^1
Glycerin (285 K)	1.19×10^{-3}	9.4×10^{-8}	5.0×10^{-4}	1.3×10^3
Water (310 K)	6.95×10^{-7}	1.52×10^{-7}	3.0×10^{-4}	8.4×10^5

Remarks: Air and especially helium yield small Ra (weak natural convection). Water produces a large Ra $\sim 10^6$, indicating strong buoyancy-driven flow. (Any handbook values close to those above will give essentially the same orders of magnitude.)

Problem 9.4: Lumped Cooling with $h = C \Delta T^{1/4}$

A solid of volume V , surface area A , density ρ , and heat capacity c cools in a quiescent fluid with ambient temperature T_∞ . The free-convection coefficient is

$$h = C [T(t) - T_\infty]^{1/4}.$$

Under the lumped-capacitance approximation ($\text{Bi} \ll 0.1$), the energy balance is

$$\rho c V \frac{dT}{dt} = -h A [T - T_\infty] = -C A [T - T_\infty]^{5/4}.$$

Let $\theta(t) = T(t) - T_\infty$. Then

$$\frac{d\theta}{dt} = -\frac{C A}{\rho c V} \theta^{5/4}.$$

Separate and integrate from $\theta_i = T_i - T_\infty$ at $t = 0$ to $\theta_f = T_f - T_\infty$ at time t :

$$\int_{\theta_i}^{\theta_f} \theta^{-5/4} d\theta = -\frac{C A}{\rho c V} \int_0^t dt \Rightarrow -4 \left[\theta^{-1/4} \right]_{\theta_i}^{\theta_f} = -\frac{C A}{\rho c V} t.$$

Hence the cooling time is

$$t = \frac{4\rho cV}{AC} \left[(T_f - T_\infty)^{-1/4} - (T_i - T_\infty)^{-1/4} \right].$$

(Equivalently, replace ρcV by mc_p .)

Validity: Use when the Biot number $\text{Bi} = \frac{hL_c}{k}$ (with $L_c = V/A$) satisfies $\text{Bi} \lesssim 0.1$.